

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 14: Equation of a Line

Topic-wise short notes with solved examples

Every straight line has an equation that all its points satisfy. This chapter covers the slope of a line, the conditions for parallel and perpendicular lines, and the different ways of writing the equation.

1 A point on a line

- The equation of a line is a relation between x and y that is satisfied by the coordinates of **every** point on the line, and by no other point.
- A point lies on a line if and only if its coordinates satisfy the equation of the line.

Example 1.1 Does the point $(2, -1)$ lie on the line $3x + 4y = 2$? Find k if $(k, 3)$ lies on this line.

SOLUTION

$3(2) + 4(-1) = 6 - 4 = 2$. The equation is satisfied, so the point lies on the line.

$3k + 4(3) = 2$ gives $3k = -10$

Yes; and $k = -\frac{10}{3}$.

2 Inclination and slope

- The **inclination** θ of a line is the angle it makes with the positive direction of the x -axis, measured anticlockwise.
- The **slope** (gradient) of a line is $m = \tan \theta$.

$$m = \tan \theta \quad m = \frac{y_2 - y_1}{x_2 - x_1} \text{ for the line through } (x_1, y_1) \text{ and } (x_2, y_2)$$

Line	Inclination	Slope
Parallel to the x -axis	0°	0
Parallel to the y -axis	90°	Not defined
Inclined at 45°	45°	1
Inclined at 60°	60°	$\sqrt{3}$

Example 2.1 Find the slope and the inclination of the line through $A(2, 3)$ and $B(5, 6)$.

SOLUTION

$$m = \frac{6 - 3}{5 - 2} = 1$$

$$\tan \theta = 1$$

Slope = 1; inclination = 45°

3 Parallel and perpendicular lines

- **Parallel** lines have equal slopes: $m_1 = m_2$.
- **Perpendicular** lines have slopes whose product is -1 : $m_1 \times m_2 = -1$, so $m_2 = -\frac{1}{m_1}$.
- Three points A, B, C are **collinear** if slope of AB = slope of BC .

Example 3.1 Show that the points $A(1, 2)$, $B(3, 6)$ and $C(6, 12)$ are collinear.

SOLUTION

$$\text{Slope of } AB = \frac{6 - 2}{3 - 1} = 2 \text{ and slope of } BC = \frac{12 - 6}{6 - 3} = 2$$

The slopes are equal and B is common, so the points are collinear.

Example 3.2 The line through $(2, 5)$ and $(4, k)$ is perpendicular to the line through $(1, 1)$ and $(5, 3)$. Find k .

SOLUTION

$$\text{Slope of the second line} = \frac{3 - 1}{5 - 1} = \frac{1}{2}, \text{ so the slope of the first must be } -2.$$

$$\frac{k - 5}{4 - 2} = -2, \text{ so } k - 5 = -4$$

$$k = 1$$

4 Forms of the equation

Form	Equation	Use it when you know
Slope-intercept	$y = mx + c$	The slope and the y -intercept c
Point-slope	$y - y_1 = m(x - x_1)$	The slope and one point
Two-point	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$	Two points

- The **y -intercept** c is the y -coordinate of the point where the line cuts the y -axis. The **x -intercept** is the x -coordinate where it cuts the x -axis.
- $x = a$ is a line parallel to the y -axis and $y = b$ is a line parallel to the x -axis.

Example 4.1 Find the equation of the line with slope 2 that passes through $(1, -3)$.

SOLUTION

$$y - (-3) = 2(x - 1)$$

$$y + 3 = 2x - 2$$

$$y = 2x - 5$$

Example 4.2 Find the equation of the line through $(1, 2)$ and $(3, 8)$.

SOLUTION

$$\text{Slope} = \frac{8 - 2}{3 - 1} = 3$$

$$y - 2 = 3(x - 1)$$

$$y = 3x - 1$$

5 Slope and intercepts from an equation

- Write the equation in the form $y = mx + c$. Then m is the slope and c is the y -intercept.
- For $ax + by + c = 0$: slope $= -\frac{a}{b}$ and y -intercept $= -\frac{c}{b}$.
- For the x -intercept put $y = 0$; for the y -intercept put $x = 0$.

Example 5.1 Find the slope, the y -intercept and the x -intercept of the line $3x + 4y = 12$.

SOLUTION

$$4y = -3x + 12, \text{ so } y = -\frac{3}{4}x + 3$$

$$\text{Put } y = 0: 3x = 12, \text{ so } x = 4$$

$$\text{Slope} = -\frac{3}{4}; y\text{-intercept} = 3; x\text{-intercept} = 4$$

6 Lines parallel or perpendicular to a given line

Example 6.1 Find the equation of the line through $(2, 1)$ that is (i) parallel (ii) perpendicular to the line $2x - 3y = 5$.

SOLUTION

$$\text{The given line is } y = \frac{2}{3}x - \frac{5}{3}, \text{ so its slope is } \frac{2}{3}.$$

$$(i) \text{ Parallel line: slope } \frac{2}{3}. y - 1 = \frac{2}{3}(x - 2), \text{ so } 3y - 3 = 2x - 4$$

$$(ii) \text{ Perpendicular line: slope } -\frac{3}{2}. y - 1 = -\frac{3}{2}(x - 2), \text{ so } 2y - 2 = -3x + 6$$

$$(i) 2x - 3y - 1 = 0 \quad (ii) 3x + 2y - 8 = 0$$

Example 6.2 Find the equation of the perpendicular bisector of the segment joining $A(2, 3)$ and $B(6, 5)$.

SOLUTION

Mid-point of $AB = (4, 4)$

Slope of $AB = \frac{5-3}{6-2} = \frac{1}{2}$, so the slope of the perpendicular bisector is -2 .

$$y - 4 = -2(x - 4)$$

$$2x + y = 12$$

7 Common mistakes

- Finding the slope as $\frac{x_2 - x_1}{y_2 - y_1}$. It is the change in y over the change in x .
- Reading the slope from $3x + 4y = 12$ as 3. First make y the subject.
- Using $m_1 m_2 = 1$ for perpendicular lines. The product is -1 .
- Forgetting that the perpendicular bisector passes through the **mid-point**.